ABSTRACT

Semiconductor manufacturing underpins nearly every modern electronic device, from mobile consumer electronics to mission-critical automotive and healthcare systems. Detecting wafer defects early during fabrication is vital to maximizing die yield, preventing downstream failures, and curtailing manufacturing overhead. However, conventional machine learning defect detection systems rely heavily on centralized data collection, requiring original equipment manufacturers (OEMs) and foundries to share proprietary wafer inspection maps—a practice strongly resisted due to intellectual property and trade secret concerns. Furthermore, state-of-the-art deep neural networks operate as opaque black boxes, providing negligible insight to process engineers as to why specific wafer patterns are flagged defective. This seminar presents a detailed review of the paper **"Demystifying Defects: Federated Learning and Explainable AI for Semiconductor Fault Detection"** by T. Patel, R. Murugan, G. Yenduri, R. H. Jhaveri, H. Snoussi, and T. Gaber, published in *IEEE Access*, Vol. 12, 2024. The paper introduces a privacy-preserving, transparent inspection framework combining decentralized federated training with post-hoc explainability.

The proposed framework distributes convolutional neural network backbones across ten client manufacturing stations, training models locally on private wafer image shards while ensuring raw data never leaves client boundaries. A central coordinator aggregates model weights via Federated Averaging (FedAvg) enhanced with calibrated Gaussian noise to enforce formal Differential Privacy against reconstruction attacks. Four deep learning architectures (ResNet152, DenseNet121, MobileNetV2, InceptionV3) were evaluated on the benchmark MixedWM38 dataset comprising over 38,000 grayscale wafer maps across 38 defect classes. The ResNet152 federated model achieves a peak test accuracy of **98.78%**, outperforming prior deformable convolutional network baselines (DCNet, 93.20%) by nearly 5.6 percentage points. An integrated Explainable AI layer combining Gradient-weighted Class Activation Mapping (Grad-CAM) and Local Interpretable Model-agnostic Explanations (LIME) produces interpretable heatmaps and super-pixel attributions that confirm the model focuses on genuine physical defect clusters rather than background noise. This approach demonstrates that high detection accuracy, privacy preservation, and model transparency can be simultaneously achieved in modern semiconductor fabrication.

Keywords : Semiconductor Manufacturing, Wafer Defect Detection, Federated Learning, Explainable AI (XAI), Differential Privacy, Grad-CAM, LIME.

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